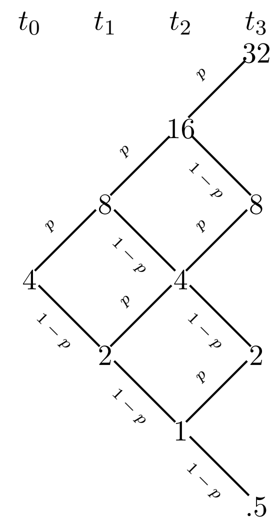
MSCF Probability, August 2018: Homework Problems, Set 2

1. **Binomial Model for Stock Prices**
   The *binomial model* for stock prices describes the evolution of prices in discrete time. Starting at a value S, the value in the next period could be uS with probability p or dS with probability $1-p$. If $S_0 = 4$, $u = 2$, and $d = 1/2$ at time t_0, then the model has the following structure:



   Let the random variable X_T be the number of upward moves in T periods, so $X_T \sim \text{Bin}(T, p)$. With this variable, we can then make probability statements about the value at time t, S_t. For instance, we can see that any path to a value of 8 at t_3 must involve two upward steps and one downward step. Consequently,

$$P(S_3 = 8) = P(X_3 = 2) = \binom{3}{2} p^2 (1-p)^1.$$

   For the example above, and using $p = .4$, find $P(S_7 \geq 32)$.

2. The number of times that a person contracts a cold in a given year is a Poisson random variable with parameter $\lambda = 5$. Suppose that a new wonder drug has just been marketed that reduces the Poisson parameter to $\lambda = 3$ for 60% of the population. For the other 40% of the population, the drug has no appreciable effects on colds. If an individual tries the drug for a year and has 2 colds in that time, how likely is it that the drug is beneficial for him or her?

3. Let X have the density function given by: $f_X(x) = 0.5e^{-|x|} \quad -\infty < x < \infty$
   Find $P(2.5 \leq |X| \leq 3.5)$.

4. The life of an electronic component in hundreds of hours (X) has an exponential distribution with *mean* of 10. Given that the component has been working for 900 hours, what is the probability that it will work for no more than 1100 hours?
   Repeat under the assumption that the lifetime in hundreds of hours is not exponential but rather uniformly distributed over (8,12).

5. Suppose that $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$, and that X and Y are independent. What is the distribution of $\min(X, Y)$?